

Supplemental tables and figures

For additional information and details on analysis please visit <https://alexkoiter.ca/terrain-attributes/>

Table S1: Description of spectral reflectance colour coefficients used as fingerprints. Reproduced from Boudreault et al. (2018)

Colour space model	Parameter	Abbreviation
RGB	Red	R
RGB	Green	G
RGB	Blue	B
CIE xyY	Chromatic Coordinate x	x
CIE xyY	Chromatic Coordinate y	y
CIE xyY	Brightness	Y
CIE XYZ	Virtual component X	X
CIE XYZ	Virtual component Z	Z
CIE LAB	Metric lightness function	L
CIE LAB	Chromatic coordinate opponent red-green scales	a^*
CIE LAB	Chromatic coordinate opponent red-green scales	b^*
CIE LUV	Chromatic coordinate opponent blue-yellow scales	u^*
CIE LUV	Chromatic oordinate opponent red-green scales	v^*
CIE LCH	CIE hue	c^*
CIE LCH	CIE chroma	h^*

Table S2: Terrain attribute descriptions

Terrain Attribute	Description
Elevation	Meters above sea level
Plan Curvature	Across slope curvature
Profile Curvature	Down slope curvature
SAGA Wetness Index	Similar to the 'Topographic Wetness Index' (TWI), but it is based on a modified catchment area calculation, which does not think of the flow as very thin film. As result it predicts for cells situated in valley floors with a small vertical distance to a channel a more realistic, higher potential soil moisture compared to the standard TWI calculation
Catchment Area	Area of upslope contributing area
Relative Slope Position	A value between 0 and 1 illustrating the position of a pixel within the landscape with values approaching 0 indicating streams to pits, and values approaching 1 indicating upper slope positions to peaks
Vertical Distance to Channel	The vertical distance to a channel network base level. The algorithm consists of two major steps: 1. Interpolation of a channel network base level elevation 2. Subtraction of this base level from the original elevations

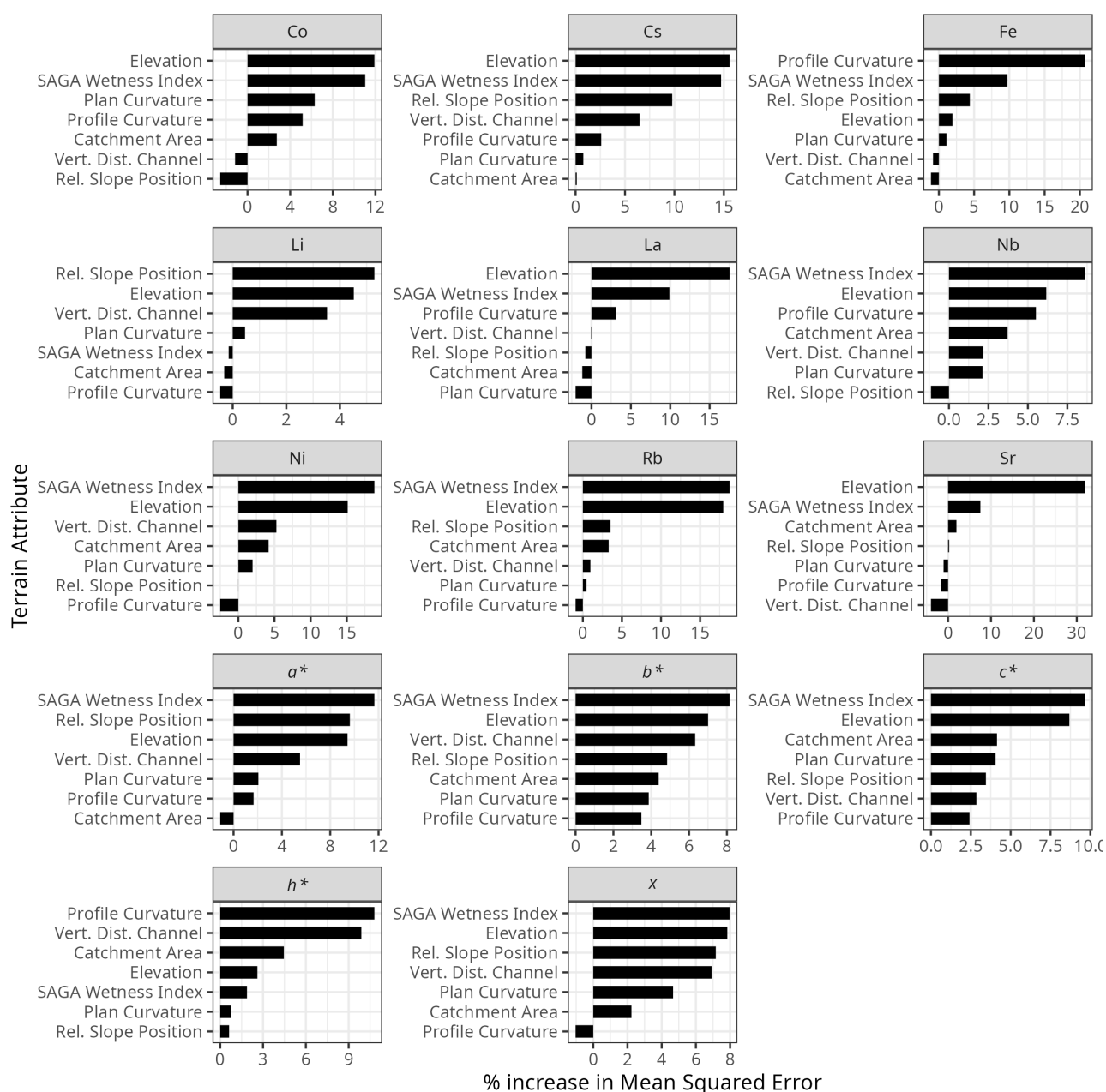


Figure S1: Variable importance, expressed as the percentage increase in Mean Squared Error, for soil colour and geochemical fingerprint properties derived from random forest models using terrain attributes as predictors at the agricultural site.

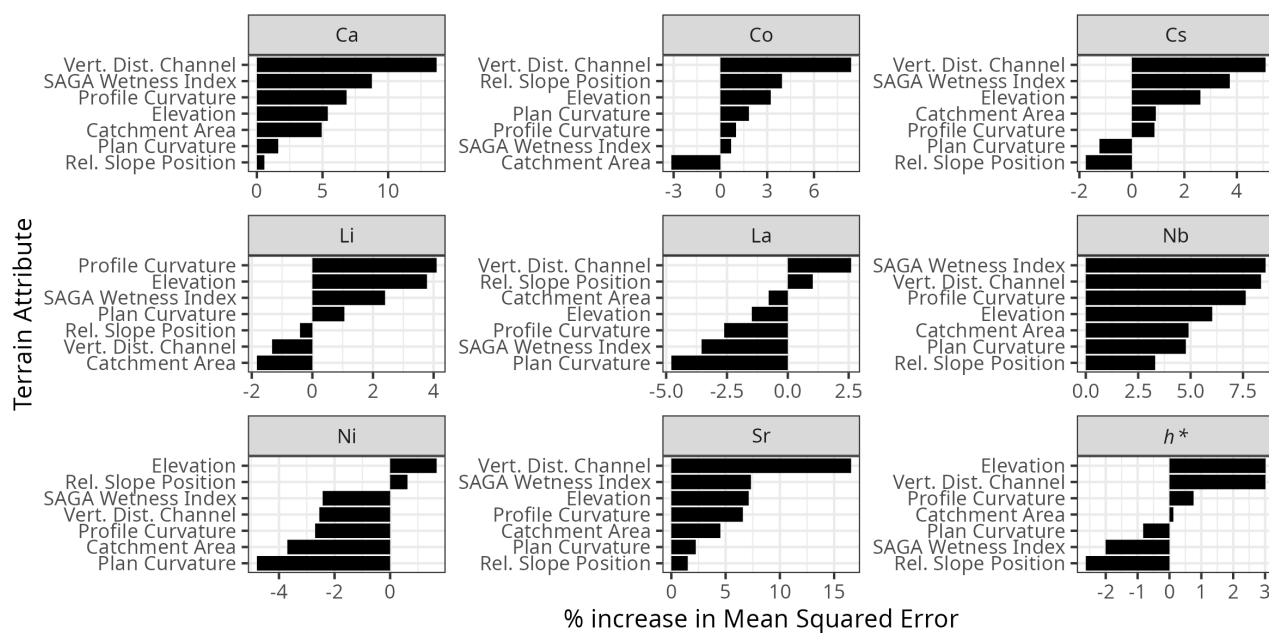


Figure S2: Variable importance, expressed as the percentage increase in Mean Squared Error, for soil colour and geochemical fingerprint properties derived from random forest models using terrain attributes as predictors at the forest site.